IMAGE PROCESSING TECHNIQUES – EXTENDED CLASS ACTIVITIES WORKSHEET

UG Level | Duration: 10 Minutes per Activity

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Activity 11: Canny Edge Detection – Stage Analysis

Arrange the following steps in correct order:

• Double Thresholding

• Gaussian Smoothing

• Non-Maximum Suppression

• Gradient Calculation

Correct Order:

1. Gaussian Smoothing

2. Gradient Calculation

3. Non-Maximum Suppression

4. Double Thresholding

Questions:

1. What is the purpose of Non-Maximum Suppression?

Non-Maximum Suppression (NMS) is used in the Canny edge detection process to make detected edges **thin and precise**. It compares the gradient magnitude of a pixel with its neighboring pixels along the gradient direction. If the pixel does not have the maximum intensity, it is suppressed and set to zero. This removes unnecessary pixels and produces **thin, well-defined edges** instead of thick or blurred edge regions.

2. Why does Canny use double thresholding?

Canny uses **double thresholding** to classify detected edges into **strong edges, weak edges, and non-edges**. A high threshold identifies pixels that are definitely edges, while a low threshold identifies weaker possible edges. Weak edges are retained when they are connected to strong edges, while isolated weak edges are removed. This helps Canny **preserve important edges while reducing false or noisy edges**.